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Solar Potential Assessment using PVsyst Software in the Northern Zone of Morocco

Brahim Belmahdi[a,]*, Abdelmajid El Bouardi[a]

[a]*ETEE, Abdelmalek Essaadi University, Faculty of Science, Tetuan, Morocco*

Abstract

The growing demand for solar photovoltaic (PV) installation systems as a renewable energy source is required in terms of accuracy to predict system efficiency. We have adopted in this paper a complete demonstration of the modelling and simulation of a 1 MW photovoltaic plant in one zone located in the north of Morocco, as a study; we will therefore use the PVsyst software for this purpose and evaluate its performance. The performance ratio and power losses such as temperature and electric power are calculated. As a result, the possibility of installing a 1 MW solar plant is examined by comparing the solar energy produced at certain sites in the northern zone of Morocco.

Keywords: solar photovoltaic; renewable energy; PVsyst; performance ratio.

1. Introduction

Photovoltaic (PV) systems are developing rapidly. As a renewable energy source, they play an increasingly important role in the world by providing a safe environment. The disadvantages of the solar power plant are the intermittency of the source, the high cost of installation and the energy conversion performance (from 12% to 29%) relatively low [1].

* Corresponding author. Tel.: +212-660-613-174.
E-mail address: belmahdi.brahim@gmail.com

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Therefore, the photovoltaic system (PV) must be designed to operate at its maximum efficiency in order to maintain the maximum power of the photovoltaic system panel, when their weather conditions are favorable.

2. Geographical and parameters coordinate of northern Moroccan

The Kingdom of Morocco has more than 5 KWh / m² / d on average and more than 3.000 KWh of sunshine a year to develop the national source of renewable energy. To achieve this goal, the world is set at 42% of energy sources by 2020; it has been increased to 52% by 2030. The major project aims to create a capacity of 2 gigawatts on five radii sun (Ourzazate, Boujdour, Ain Bni Methar, Foum Al Oued and Sebkhat Tah) by 2020, including photovoltaics and concentrated solar energy [2,3].

Table 1. Geographical coordinates details.

	Cities	Altitude	Longitude	Latitude
1	Tetuan	17	-5.38	35.58
2	Tangier	21	-5.9	35.73
3	Larache	49	-6.13	35.18
4	Al-Hoceima	100	-3.92	35.23

Morocco is the only African country to exchange electricity produced from renewable energy sources with Europe. The strategic position of Morocco occupies an important place, it is located in the north of Africa, and it is surrounded by the Mediterranean Sea in the north [4, 5]. Table.1 describes the geographical coordinates of four sites taken into account and are considered as input of the proposed study.

3. Methodology

The photovoltaic system contains solar panels, inverters, power conditioning unit and network connected equipment. In addition, the cost of these components decreases continuously. The PV system is based on annual amount of energy produced, specific product and performance ratio [6]. For this purpose, we compare in this study the efficiency of photovoltaic for both fixed tilted plan and seasonal adjustment using PVsyst software package ; whereas, the energy produced is compared with three places located in northern zone such as Tangier, Larache and Al-hoceima.

3.1. System Components Summary

Solar panel depends on many factors such as modules, inverters quality, geographical coordinates, inclination and orientation of PV panels [7]. Tetuan city is taken into account in this study (Latitude 35.56° N, Longitude 5.36 °E); the measured data such as monthly global solar radiation (G_h), monthly diffuse (G_d), average Temperature ($T_{average}$) and wind speed (W_s), are presented in Table 2. The previous parameters are provided by the energetic laboratory of the faculty of science in Tetuan city and considered as an input data. PVsyst software employed Perez Model for estimating incident solar irradiation on titled surface plane [8].

Table 2. Measured data of Tetuan city.

Interval beginning	Gh (KWh/m²)	Gd (KWh/m²)	Taverage (°C)	Ws (m /s)
January	102.3	30.88	13.5	4.1
February	103.6	39.66	14.1	4.6
March	136.4	61.21	15.9	4.9
April	171	63.55	17.1	5

May	195.3	69.69	19.8	4.9
June	228	61.33	23.2	4.9
July	226.3	67.71	26.1	4.7
August	201.5	66.31	26	4.5
September	156	57.3	23.6	4.3
October	133.3	46.64	20.7	3.7
November	114	28.72	16.4	4.1
December	96.1	25.88	14.3	4.4
Year	1863.8	618.88	19.26	5

3.2. *Orientation and inclination*

The optimization of the PV system depends on two characteristics; the first is the orientation according to the solar trajectory. The hour angle at the local solar noon is mentioned in Fig.1 (1: 22 June, 2: 22 May etc...) . The second is inclination; both aim to achieve maximum solar radiation. The Fig.2 describes two structure studies such as:

-Tilt panel fixed: the angle of inclination optimized is 32 ° and the azimuth angle is 0 °.

-Seasonal tilt adjustment: the angle of inclination is titled with 15° for summer and 48° for winter.

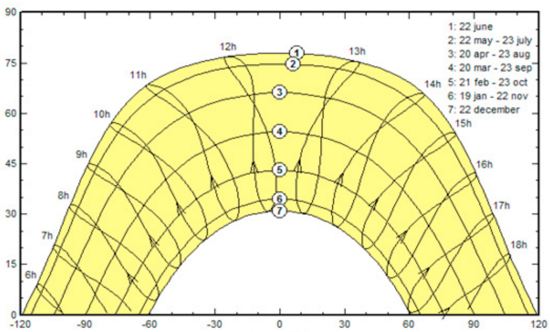

Fig. 1. Sun's path diagram for Tetuan city, northern Morocco.

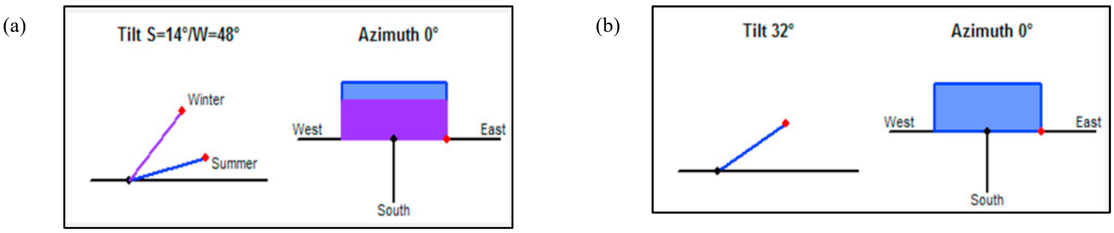

Fig. 2. (a) Seasonal adjustment tilt; (b) Fixed tilt panel of Tetuan city.

3.3. *Photovoltaic description*

The construction of grid connected PV panel system for our case study summarized with specific details in Table 3. The proposed configuration generates 1MW of power, with 3924 silicon-polycrystalline modules. The area concerned is 7715 m². A total of 10 inverters of 100 KW are implemented for grid compatibly in Tetuan site.

Table 3. Characteristics operating of the proposed configuration.

Proposed configuration	
Module Type	Standard
Technology	Si-Poly
Nominal Power	1000 KWp
Load Profile	Grid-connected
No. Of Modules	3924
Module Area	7715m²
No.of Inverters	10
PV module	255 Wp 30V
Vmpp(50°C)	31
Voc(12°C)	45.2
Number of Inverter	10
type of inverter	100KW 300-600 V 50-60Hz
conditional operating	
Vmpp(60°C)	363 V
Vmpp(20°C)	431 V
Voc(10°C)	578 V
plan Irradiance	1000 W/m²
Impp(STC)	2378 A
Isc(STC)	2632 A
Max operating power	971KW
Array nominal Power	1001KWp

3.4. PV arrays efficiency

The efficiency of the PV power (η_{Systme}) is the ratio between the energy injected to the grid (E_{Actual}) and the total radiation (G_{Total}) existed at the collector plane.It can be calculated as given in equation (1) :

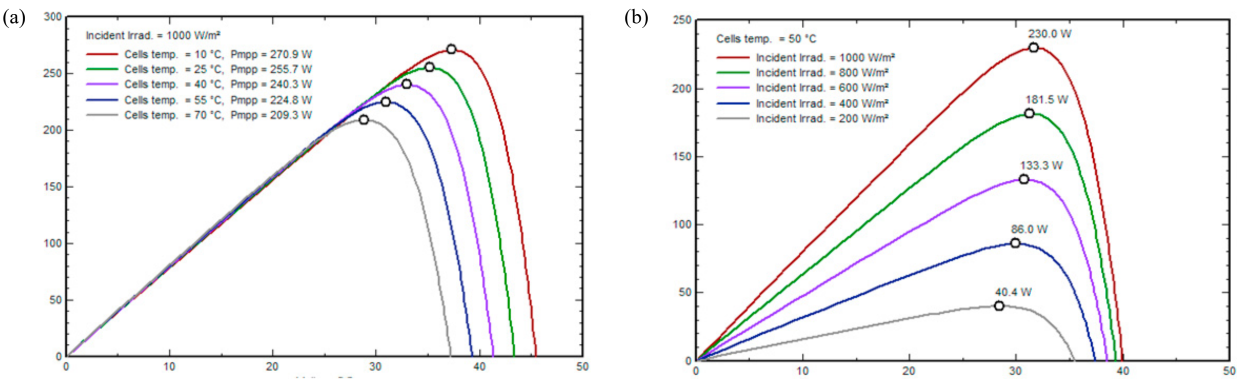

Fig. 3. (a) Comparison of curve incident irradiation; (b) Cell temperature of module SF260-36-P255.

$$\eta_{System} = \frac{E_{Actual}}{G_{Total}} \qquad (1)$$

The optimal sizing is Si-Poly SF260-36-P255 solar fun, the peak power output is 260 W. The presented Fig.3 shows the various comparisons between different curve incidence irradiation and cell temperature, as we can see, the increasing of incident irradiation and cell temperature are related by the increasing of voltage.

3.5. Inverter Efficiency

Inverter efficiency(η_{Inverter}) is the ratio of AC power (P_{AC}) output to the (P_{DC}) input of the inverter showed in the equation (2) :

$$\eta_{Inverter} = \frac{P_{AC}}{P_{DC}}. \quad (2)$$

The inverter implementation is the power gate AE from Satcom company, the nominal power 100 KW. The input voltage is close to 300-600v with maximum efficiency of 97.0 % presented in Fig.4

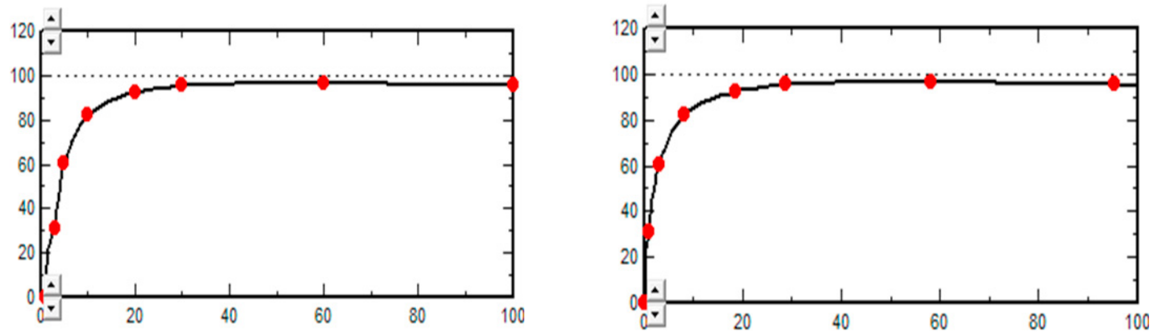

Fig. 4. (a) Input Ac; (b) Output Dc efficiencies curve of Inverter 100KW 300-600 V.

4. Result and Discussion

The experimental results based on metrological data using PVsyst software are achieved; in the meantime, the comparative study of two configurations and simulations shows that they are performed on the same characteristic of the proposed model (weather conditions and geographical climate) from January 2015 to December 2015.

Table 4. Main result between fixed and seasonal adjustment tilts.

	Fixed Tilt	Seasonal Adjustment Tilt
Global horizontal irradiation (KWh/m²)	1227.4	1227.4
Global Incident irradiation (KWh/m²)	1416.4	1348.3
Ambient Temperature (°C)	9.31	9.31
Efficiency global, corr.for IAM and shadings (KWh/m²)	1379.2	1306.0
Energy injected into grid (MWh)	1096.1	1038.2
Energy at the output of the array (MWh)	10.70	10.66
Performance ratio (%)	0.773	0.769

Table 4 represents an assessment main result of fixed tilt panel and seasonal adjustment tilt. The Array energy of the PV power plant for fixed and seasonal adjustments is close to 10.70 MWh. The Output of the inverter is the amount of energy injected to the grid which is respectively 1096.1MWh and 1038.2 MWh.

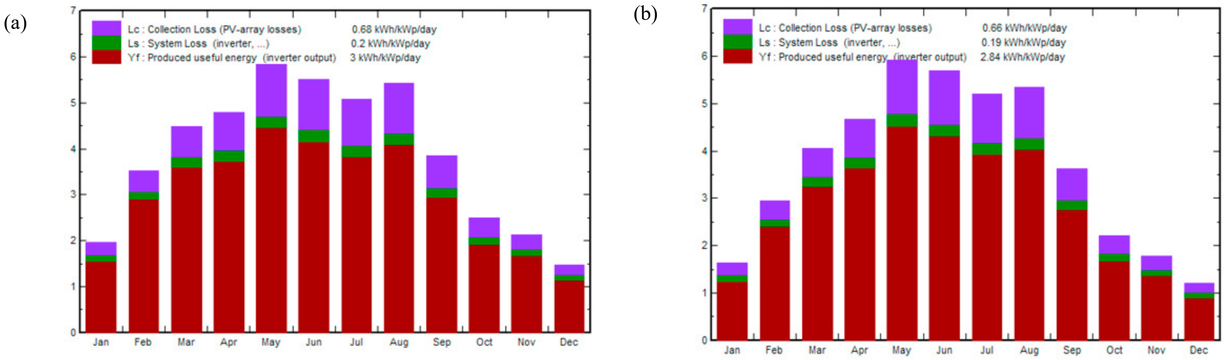

Fig. 5. (a) Energy output throughout the year (KWh/KWp/day): Titled; (b) Energy output throughout the year (KWh/KWp/day): seasonal Adjustments.

The performance ratio is the ratio of the final PV system yield (Y_f) and the reference yield (Y_r). It represents the entire losses of the system, were the energy from Dc rating to Ac output. Due to the overall component of the system (transformer, inverter, internal network etc.…) [9-12], the PR can be calculated as given in equation.3:

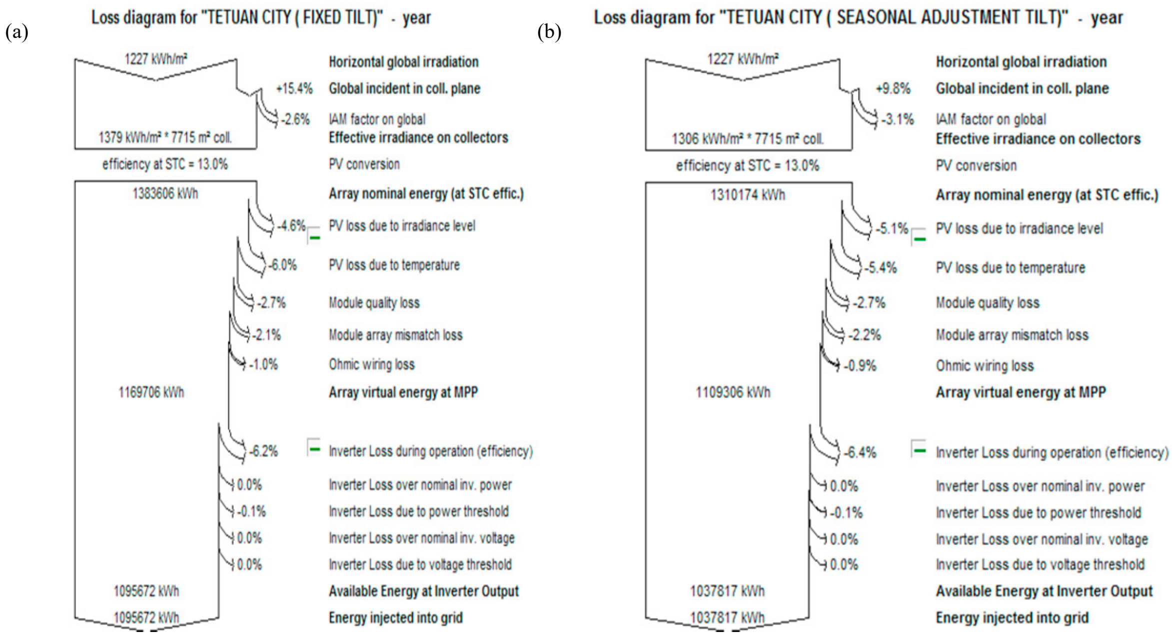

Fig. 6. Losses diagram of (a) fixed tilt; (b) seasonal adjustment tilt.

$$PR = \frac{Y_f}{Y_r}. \qquad (3)$$

The performance ratio for the coming 25 years will be close to 77.3 %.

Concerning the normalized production on fixed tilt panels and seasonal adjustments in Tetuan city, it is noted from Fig.5 (a)and (b), which its monthly average throughout the year of collection loss, system loss and the produced energy yield close to 0.68 (KWh/KWp/day), 0.2 (KWh/KWp/day), and 3 (KWh/KWp/day) respectively for fixed tilt panels, were the value of seasonal adjustments is 0.66 (KWh/KWp/day), 0.19 (KWh/KWp/day) and

2.84 (KWh/KWp/day). The following Fig. 6 takes into account the PV panels system by recognizing the main source of the losses. As we can see, the global incident irradiation on seasonal adjustment is reasonably higher than the fixed tilt with a difference 2%. The Incidence Angle Modifier (IAM) is almost + 15.4% for fixed tilt and + 9.8 % for seasonal adjustment tilt. The main losses are resulted from the array losses which are close to 15.3%. The inverter losses are reached to - 6.2 % for fixed and - 6.4 % for seasonal adjustment tilts were the energy injected to the grid for seasonal and fixed tilt is 1037 MWh and 1095 MWh respectively with efficiency of the system equivalent to 13.0 %.

5. Solar power assessment at certain cities in the northerner zone

Table 5 tabulates the assessment of several performances indicators adopted in this purpose study with three cities located in the northern zone in Morocco. Referred to all parameters indicator cited in this analysis. Fixed tilts perform better than seasonal adjustments; at the same time, the efficiency of the PV array for the fixed tilts is higher than seasonal adjustments. Comparing the energy injected to the grid for fixed tilts of Tetuan, Tangier, Laarache and Al-hoceima cities is 1096.1 MWh, 1225.9 MWh, 1203.4 MWh, and 1191.6 MWh. Whereas, the values of seasonal adjustments are respectively 1038.2 MWh, 1174.5 MWh, 1148.9 MWh, and 1139.9 MWh. The performance ratio values of the four cities are found among 76.9% and 77.9 %.

Table 5. Comparative study of four cities located in northern zone in Morocco.

	Tetuan city		Tangier city		Larache city		Al-hoceima city	
	Fixed Tilt	Seasonal Adjustment Tilt	Fixed Tilt	Seasonal Adjustment Tilt	Fixed Tilt	Seasonal Adjustment Tilt	Fixed Tilt	Seasonal Adjustment Tilt
Global horizontal irradiation (KWh/m²)	1227.4	1227.4	1337.7	1357.7	1331.6	1351.6	1277	1297
Global Incident irradiation (KWh/m²)	1416.4	1348.3	1573.7	1513	1550.6	1486.3	1524.8	1462.4
Ambient Temperature (°C)	9.31	9.31	8.13	8.13	8.77	8.77	7.84	7.84
Efficiency global, corr. for IAM and shadings (KWh/m²)	1379.2	1306.0	1532.6	1466.6	1510.4	1440.4	1484.5	1418
Energy injected into grid (MWh)	1096.1	1038.2	1225.9	1174.5	1203.4	1148.9	1191.6	1139.7

6. Conclusion

In this paper, we studied the feasibility and viability of installing 1MW photovoltaic grid connected at various cities in northern Moroccan zone. The operated result of implementing PVsyst software was achieved at an optimal angle of 32° for fixed tilt and 15° for summer and 48° for winter on seasonal adjustment tilts. The analysis performance of two configurations was performed. As a result, the fixed tilt panel is better than seasonal adjustment tilt with close performance ratio of 77.3%. The energy production in Tetuan city is 1096.1 MWh/year as compared with other locations in the same zone. The obtained result shows that northern Morocco has great solar potential due to its strategic location.

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